

Statistics for Social Science Research

Lecture 2

August 13, 2026

Topics for today

- Frequency distribution (Chapter 2, Statistics for the Behavioral Sciences)
- Central tendency (Chapter 3, Statistics for the Behavioral Sciences)

(The slides in this power point are partly taken from the Power Point Lecture Slides to accompany the textbook “Essentials of Statistics for the Behavioral Sciences” by Frederick J Gravetter and Larry B. Wallnau.)

Tools You Will Need

- **Proportions** (math review, Appendix A)
 - Fractions
 - Decimals
 - Percentages
- **Scales of measurement** (Chapter 1)
 - Nominal, ordinal, interval, and ratio
 - Continuous and discrete variables (Chapter 1)
- **Real limits** (Chapter 1)

Revision:

Discrete and Continuous Variables

- **Discrete variable**

- Has separate, indivisible categories
- No values can exist between two neighboring categories

- **Continuous variable**

- Have an infinite number of possible values between any two observed values
- Every interval is divisible into an infinite number of equal parts

2.1 Frequency Distributions

- **A frequency distribution is**
 - An organized tabulation
 - Showing the number of individuals located in each category on the scale of measurement
- Can be either a **table** or a **graph**
- Always shows
 - The categories that make up the scale
 - The frequency, or number of individuals, in each category

2.2 Frequency Distribution Tables

- Structure of Frequency Distribution Table
 - Categories in a column (often ordered from highest to lowest but could be reversed)
 - Frequency count next to category
 - Define f = frequency, X = score, N = total number of individuals (observations)
- $\Sigma f = N$
- To compute ΣX from a table
 - Convert table back to original scores or
 - Compute ΣfX

Proportions and Percentages

Proportions

- Measures the fraction of the total group that is associated with each score

- $proportion = p = \frac{f}{N}$
- Called ***relative frequencies*** because they describe the frequency (f) *in relation to the total number* (N)

Percentages

- Expresses relative frequency out of 100
- $percentage = p(100) = \frac{f}{N}(100)$
- Can be included as a separate column in a frequency distribution table

Example 2.3

Frequency, Proportion and Percent

X	f	p = f/N	percent = p(100)
5	1	$1/10 = .10$	10%
4	2	$2/10 = .20$	20%
3	3	$3/10 = .30$	30%
2	3	$3/10 = .30$	30%
1	1	$1/10 = .10$	10%

Learning Check

- Use the Frequency Distribution Table to determine how many subjects were in the study

<u>x</u>	<u>f</u>
5	2
4	4
3	1
2	0
1	3

A

• 10

B

• 15

C

• 33

D

• Impossible to determine

Learning Check

- For the frequency distribution shown, is each of these statements True or False?

<u>x</u>	<u>f</u>
5	2
4	4
3	1
2	0
1	3

T/F

- More than 50% of the individuals scored above 3

T/F

- The proportion of scores in the lowest category was $p = 3$

Grouped Frequency Distribution Tables

- If the number of categories is very large they are combined (grouped) to make the table easier to understand
- However, information is lost when categories are grouped
 - Individual scores cannot be retrieved
 - The wider the grouping interval, the more information is lost

“Rules” for Constructing Grouped Frequency Distributions

- Requirements (Mandatory Guidelines)
 - All intervals must be the same width
 - Make the bottom (low) score in each interval a multiple of the interval width
- “Rules of Thumb” (Suggested Guidelines)
 - Ten or fewer class intervals is typical (but use good judgment for the specific situation)
 - Choose a “simple” number for interval width

TABLE 2.2

This grouped frequency distribution table shows the data from Example 2.4. The original scores range from a high of $X = 94$ to a low of $X = 53$. This range has been divided into 9 intervals with each interval exactly 5 points wide. The frequency column (f) lists the number of individuals with scores in each of the class intervals.

X	f
90–94	3
85–89	4
80–84	5
75–79	4
70–74	3
65–69	1
60–64	3
55–59	1
50–54	1

Discrete Variables in Frequency *or* Grouped Distributions

- Constructing either frequency distributions or grouped frequency distributions for ***discrete variables*** is uncomplicated
 - Individuals with the same recorded score had precisely the same measurements
 - The score is an ***exact*** score

Continuous Variables in Frequency Distributions

- Constructing frequency distributions for ***continuous*** variables requires understanding that a score actually represents an interval
 - A given “score” actually could have been any value within the score’s ***real limits***
 - The recorded value was rounded off to the middle value between the score’s real limits
 - Individuals with the same ***recorded score*** probably differed slightly in their actual performance

Continuous Variables in Frequency Distributions

- Constructing grouped frequency distributions for ***continuous variables*** also requires understanding that a score actually represents an interval
- Consequently, grouping several scores actually requires grouping several intervals
- Apparent limits of the (grouped) class interval are always ***one unit*** smaller than the real limits of the (grouped) class interval. (Why?)

Learning Check

- A Grouped Frequency Distribution table has categories 0-9, 10-19, 20-29, and 30-39. What is the width of the interval 20-29?

A

• 9 points

B

• 9.5 points

C

• 10 points

D

• 10.5 points

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- You can determine how many individuals had each score from a Frequency Distribution Table

T/F

- You can determine how many individuals had each score from a Grouped Frequency Distribution

2.3 Frequency Distribution Graphs

- Pictures of the data organized in tables
 - All have two axes
 - X-axis typically has categories of measurement scale increasing left to right
 - Y-axis typically has frequencies increasing bottom to top
- General principles
 - Both axes should have value 0 where they meet
 - Height should be about $\frac{2}{3}$ to $\frac{3}{4}$ of length

Data Graphing Questions

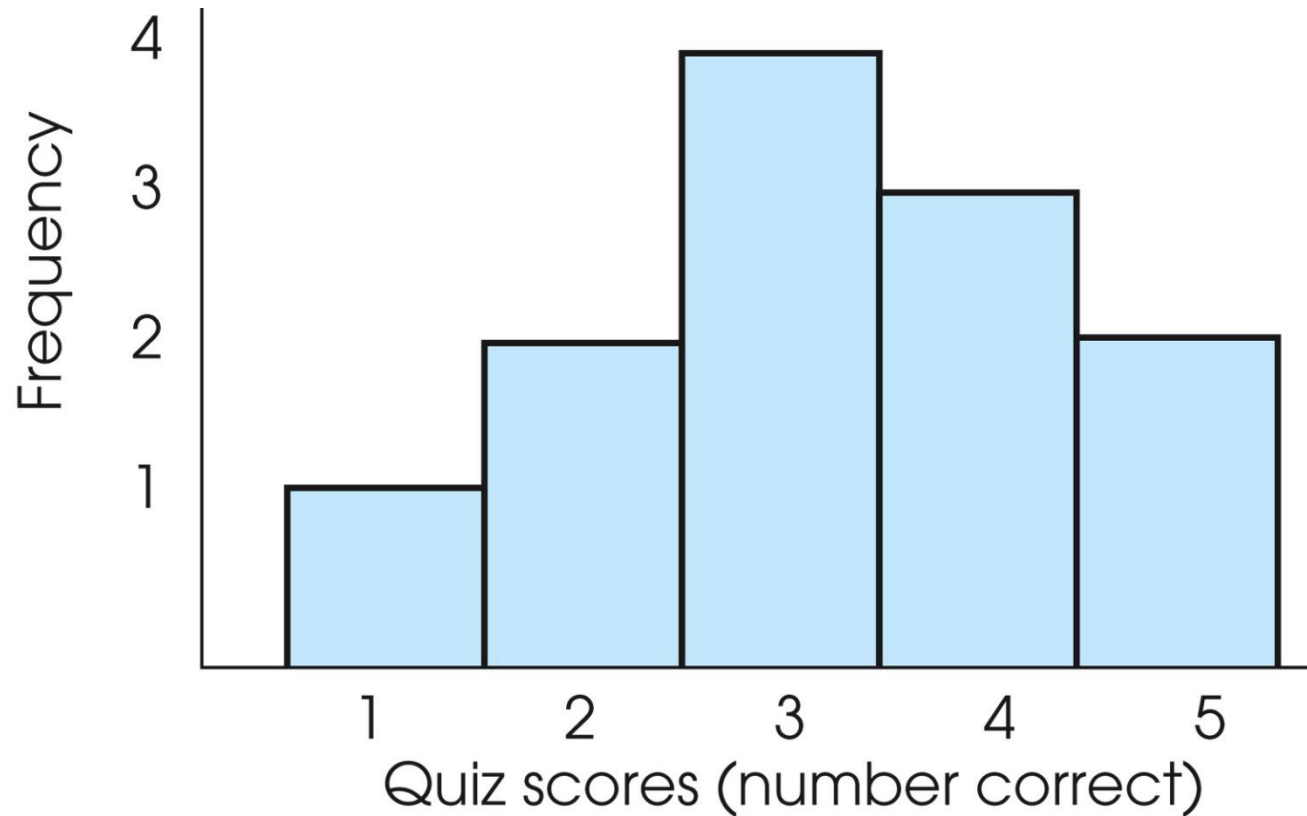
- Level of measurement? (nominal; ordinal; interval; or ratio)
- Discrete or continuous data?
- Describing samples or populations?

The answers to these questions determine which is the appropriate graph

Frequency Distribution Histogram

- Requires numeric scores (interval or ratio)
- Represent ***all*** scores on X-axis from minimum thru maximum observed data values
- Include ***all*** scores with frequency of zero
- Draw bars above each score (interval)
 - Height of bar corresponds to frequency
 - Width of bar corresponds to score real limits (or one-half score unit above/below discrete scores)

Figure 2.1
Frequency Distribution Histogram



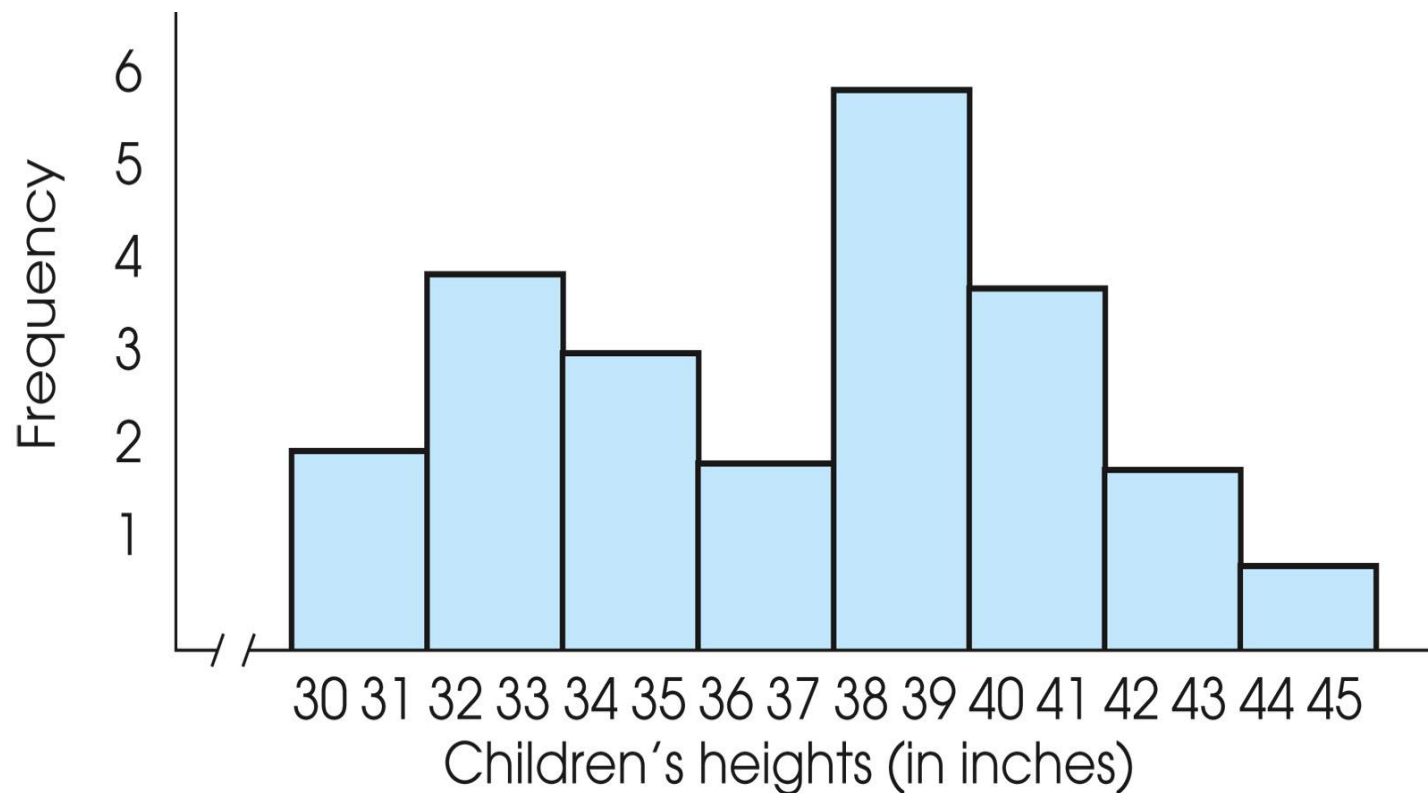
X	f
5	2
4	3
3	4
2	2
1	1

Grouped Frequency Distribution Histogram

Same requirements as for frequency distribution histogram ***except:***

- Draw bars above each (grouped) class interval
 - Bar width is the *class interval* real limits
 - Consequence? Apparent limits are extended out one-half score unit at each end of the interval

Figure 2.2
Grouped Frequency Distribution Histogram

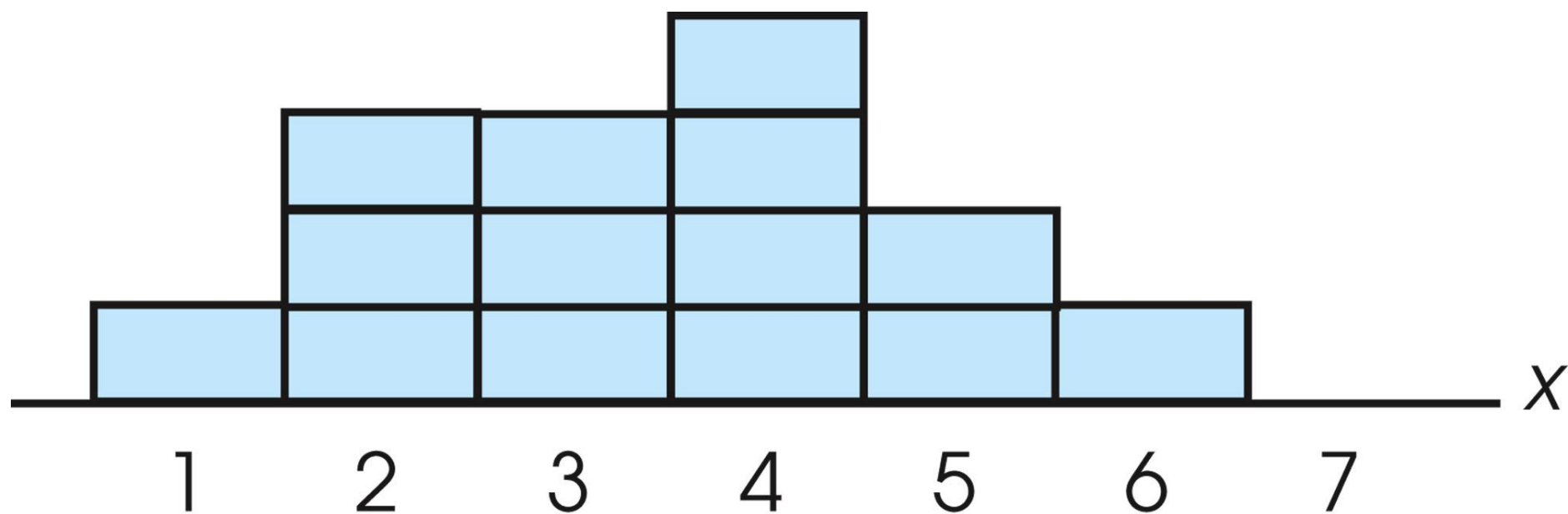


X	f
44-45	1
42-43	2
40-41	4
38-39	6
36-37	2
34-35	3
32-33	4
30-31	2

Block Histogram

- A histogram can be made a “block” histogram
- Create a bar of the correct height by drawing a stack of blocks
- Each block represents one per case
- Therefore, block histograms show the frequency count in each bar

Figure 2.3
Frequency Distribution Block Histogram



Frequency Distribution Polygons

- List all numeric scores on the X-axis
 - Include those with a frequency of $f = 0$
- Draw a dot above the center of each interval
 - Height of dot corresponds to frequency
 - Connect the dots with a continuous line
 - Close the polygon with lines to the $Y = 0$ point
- Can also be used with grouped frequency distribution data

Figure 2.4
Frequency Distribution Polygon

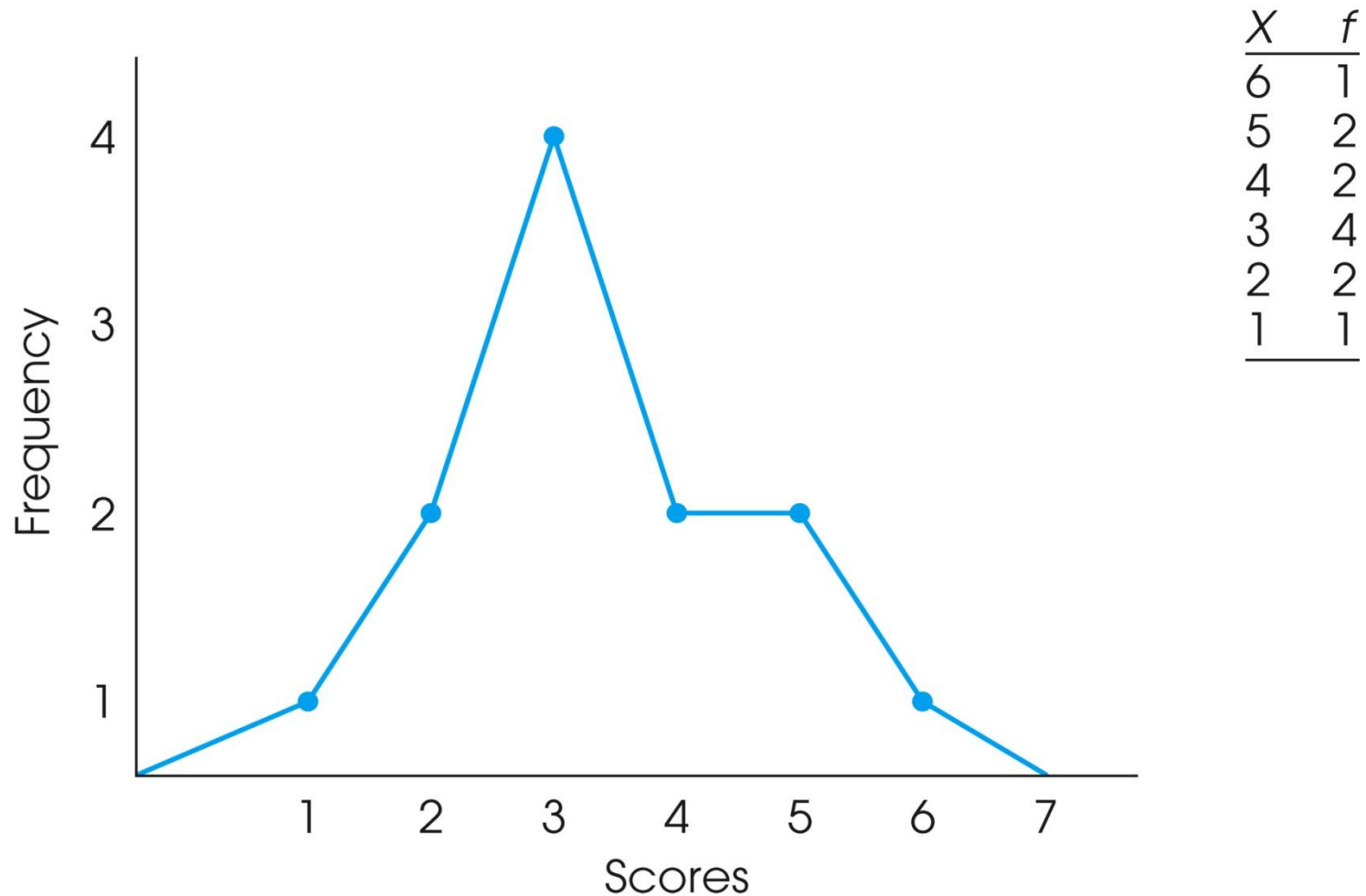
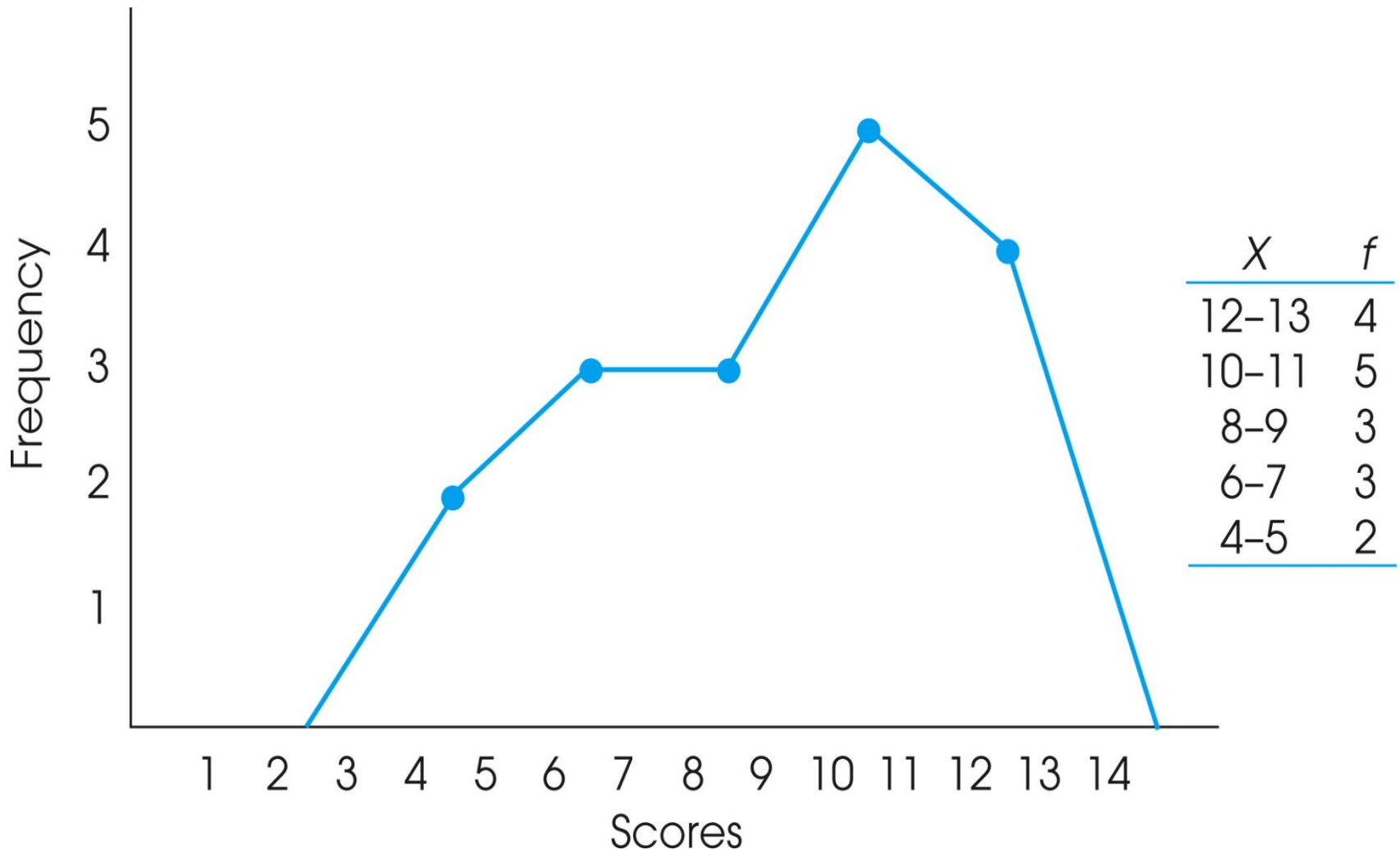


Figure 2.5

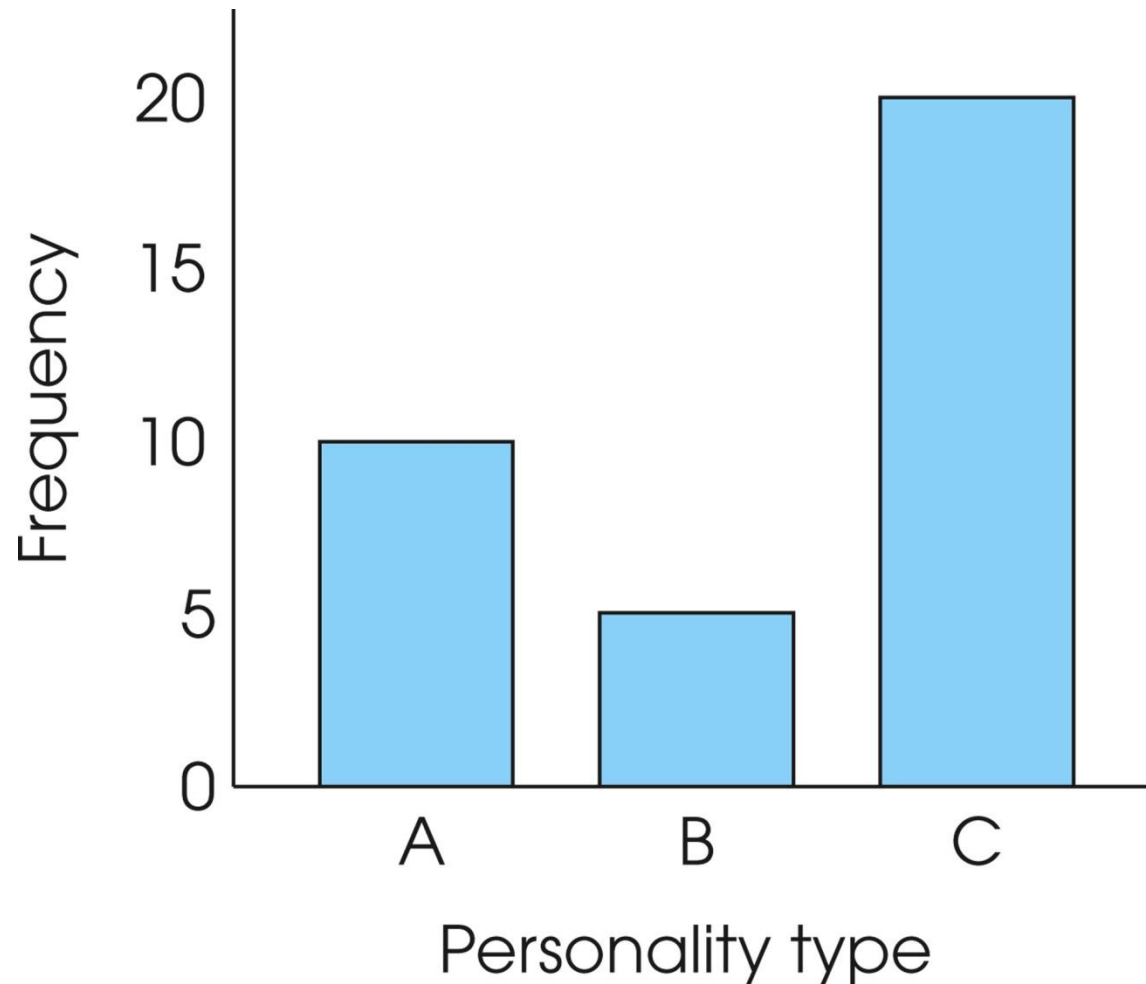
Grouped Data Frequency Distribution Polygon



Graphs for Nominal or Ordinal Data

- For non-numerical scores (nominal and ordinal data), use a **bar graph**
 - Similar to a histogram
 - Spaces between adjacent bars indicates discrete categories
 - without a particular order (nominal)
 - non-measurable width (ordinal)

Figure 2.6 - Bar graph



Population Distribution Graphs

- When population is small, scores for each member are used to make a histogram
- When population is large, scores for each member are not possible
 - Graphs based on relative frequency are used
 - Graphs use smooth curves to indicate exact scores were not used
- Normal
 - Symmetric with greatest frequency in the middle
 - Common structure in data for many variables

Figure 2.7
Bar Graph of Relative Frequencies

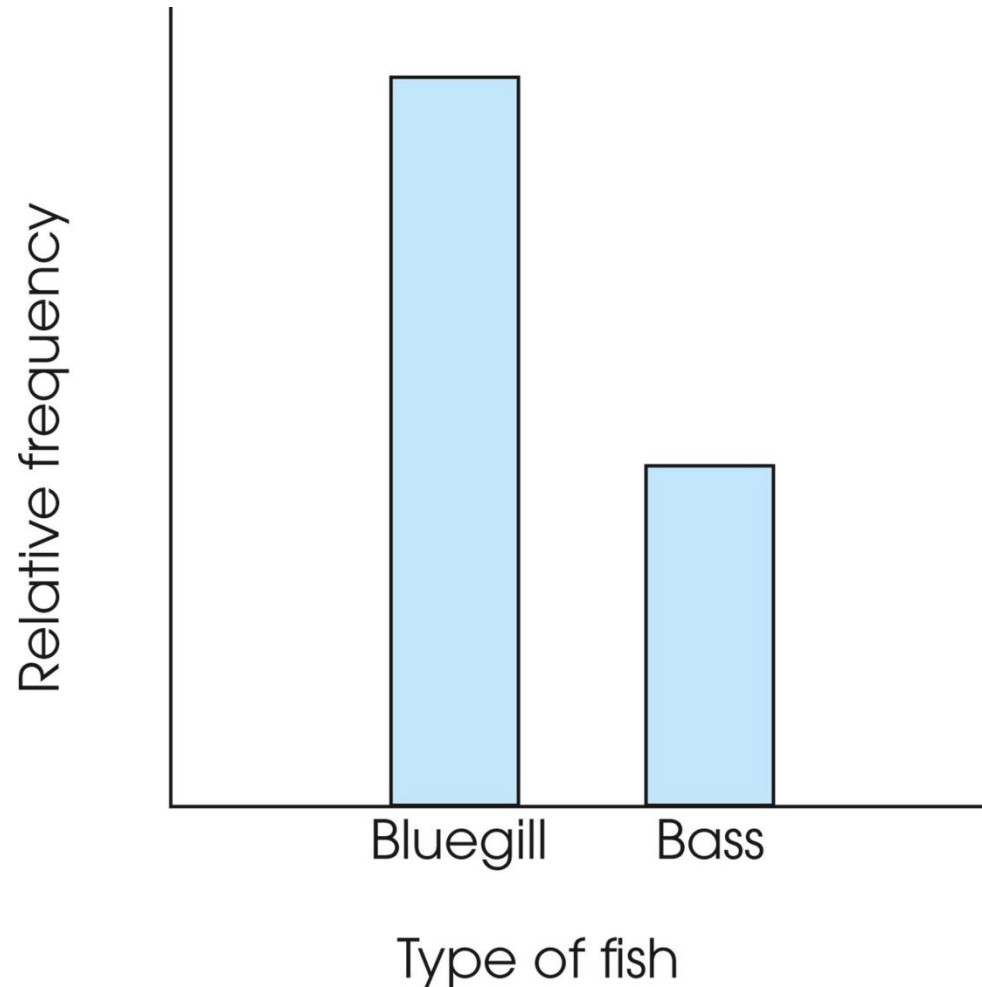
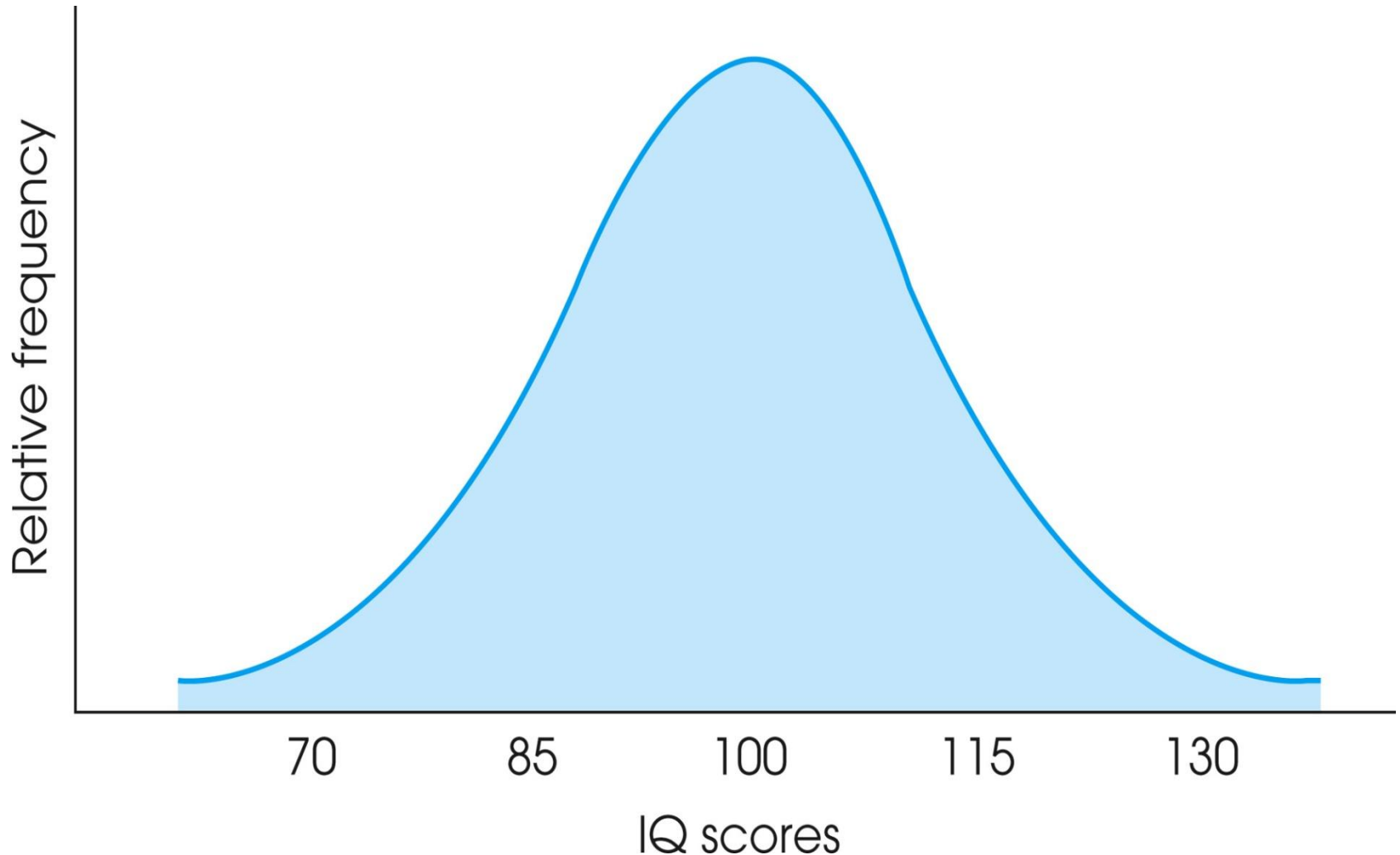
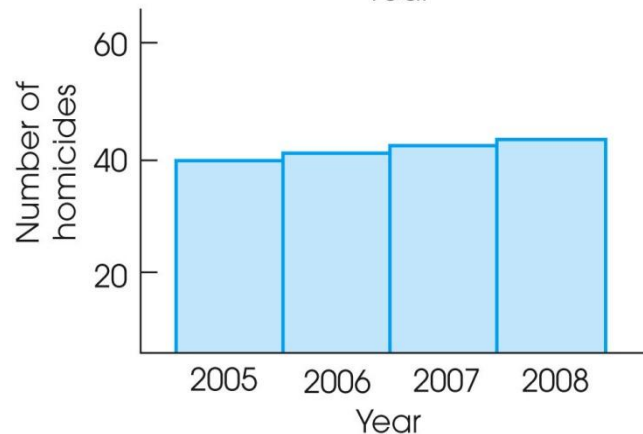
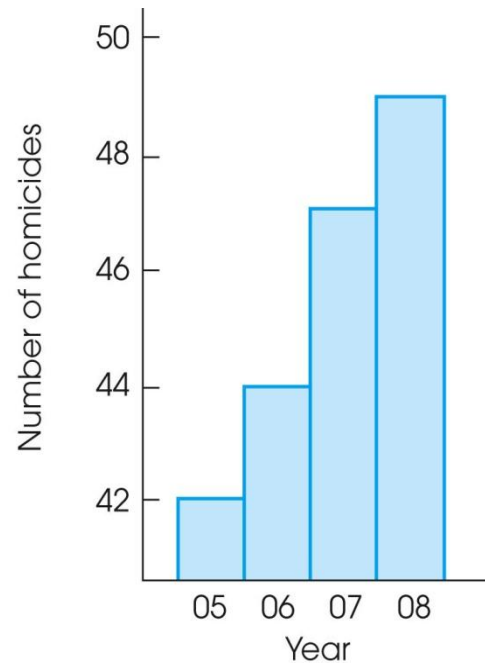


Figure 2.8 – IQ Population Distribution
Shown as a Normal Curve



Box 2.1 - Figure 2.9

Use and Misuse of Graphs

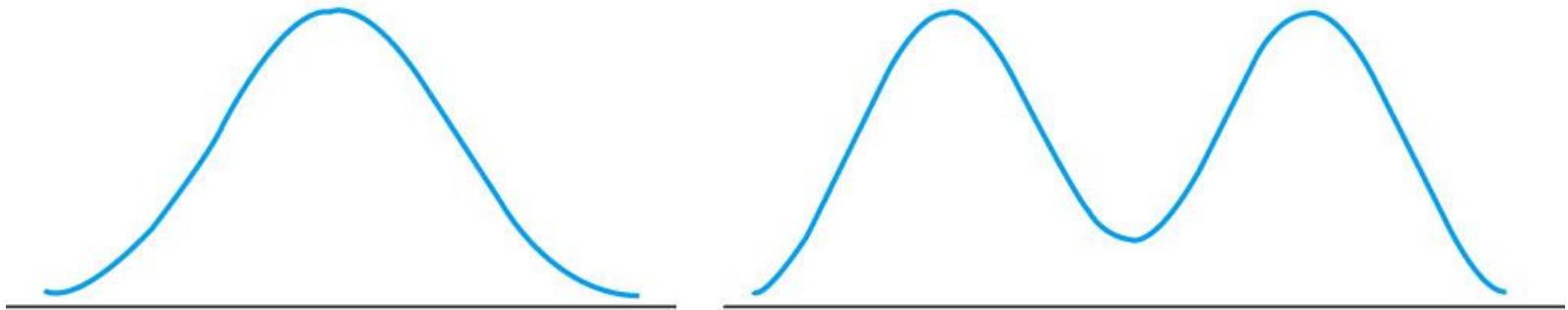


2.4 Frequency Distribution Shape

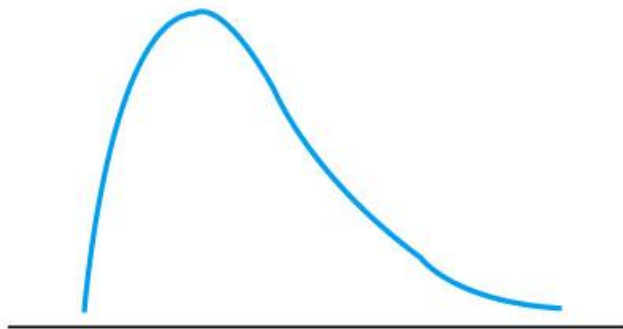
- Researchers describe a distribution's shape in words rather than drawing it
- Symmetrical distribution: each side is a mirror image of the other
- Skewed distribution: scores pile up on one side and taper off in a **tail** on the other
 - Tail on the right (high scores) = positive skew
 - Tail on the left (low scores) = negative skew

Figure 2.10 - Distribution Shapes

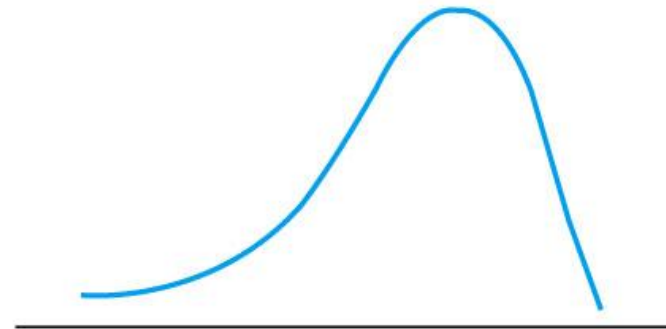
Symmetrical distributions



Skewed distributions



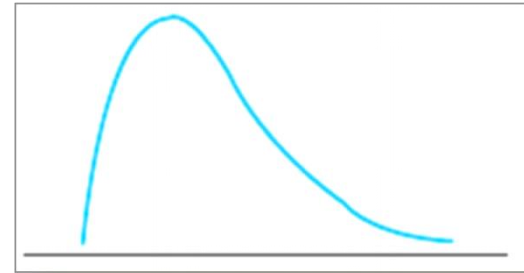
Positive skew



Negative skew

Learning Check

- What is the shape of this distribution?



A

- Symmetrical

B

- Negatively skewed

C

- Positively skewed

D

- Discrete

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- It would be correct to use a histogram to graph parental marital status data (single, married, divorced...) from a treatment center for children

T/F

- It would be correct to use a histogram to graph the time children spent playing with other children from data collected in children's treatment center

Chapter 3

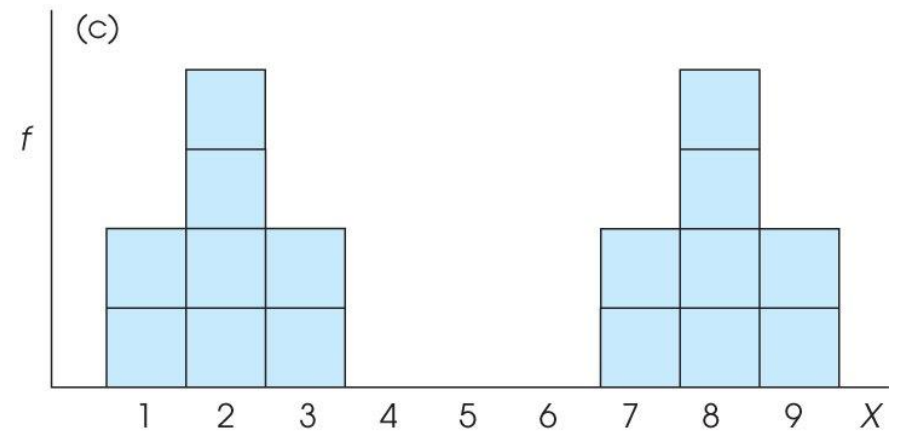
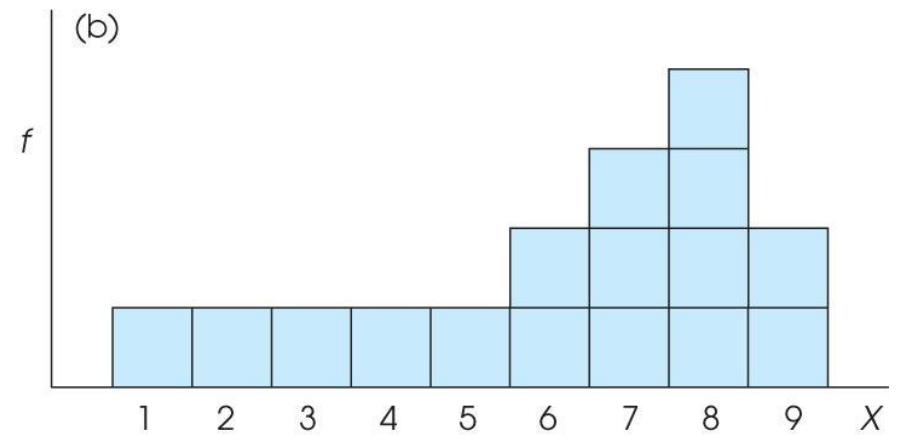
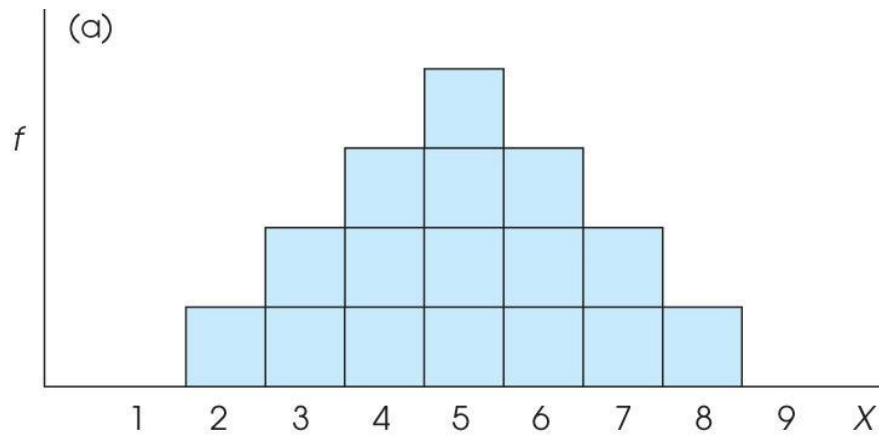
Measures of Central Tendency

3.1 Defining Central Tendency

- Central tendency
 - A statistical measure
 - A single score to define the center of a distribution
- Purpose: find the single score that is most typical or best represents the entire group

Figure 3.1

Locate Each Distribution “Center”



Central Tendency Measures

- Figure 3.1 shows that no single concept of central tendency is always the “best”
- Different distribution shapes require different conceptualizations of “center”
- Choose the one which best represents the scores in a specific situation

3.2 The Mean

- The mean is the sum of all the scores divided by the number of scores in the data.

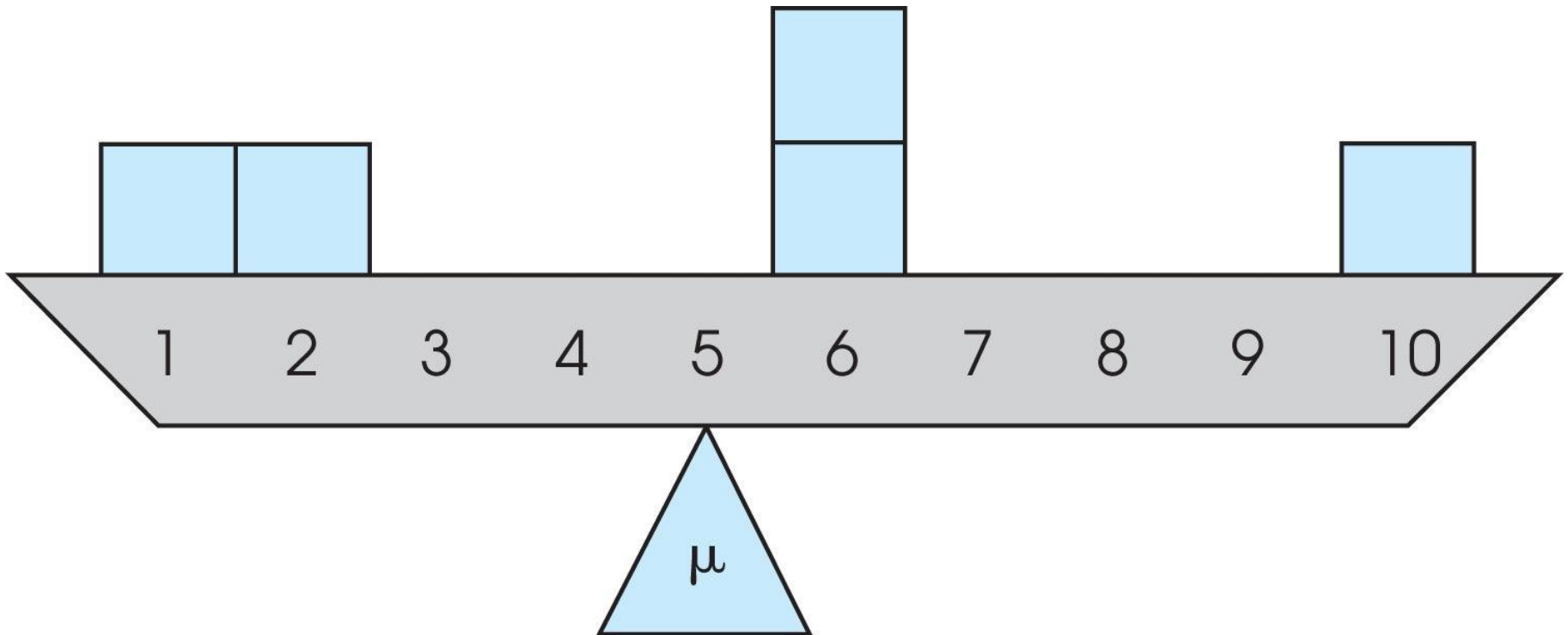
- Population:
$$\mu = \frac{\sum X}{N}$$

- Sample:
$$M = \frac{\sum X}{n}$$

The Mean: Three Definitions

- Sum of the scores divided by the number of scores in the data
- The amount each individual receives when the total is divided equally among all the individuals in the distribution
- The balance point for the distribution

Figure 3.2
Mean as Balance Point



The Weighted Mean

- Combine two sets of scores
- Three steps:
 - Determine the combined sum of all the scores
 - Determine the combined number of scores
 - Divide the sum of scores by the total number of scores

$$\text{Overall Mean} = M = \frac{\sum X_1 + \sum X_2}{n_1 + n_2}$$

Computing the Mean from a Frequency Distribution Table

Quiz Score (X)	f	fX
10	1	10
9	2	18
8	4	32
7	0	0
6	1	6
Total	$n = \Sigma f = 8$	$\Sigma fX = 66$
$M = \Sigma X / n = 66/8 = 8.25$		

Learning Check

A sample of $n = 12$ scores has a mean of $M = 8$.
What is the value of ΣX for this sample?

A

• $\Sigma X = 1.5$

B

• $\Sigma X = 4$

C

• $\Sigma X = 20$

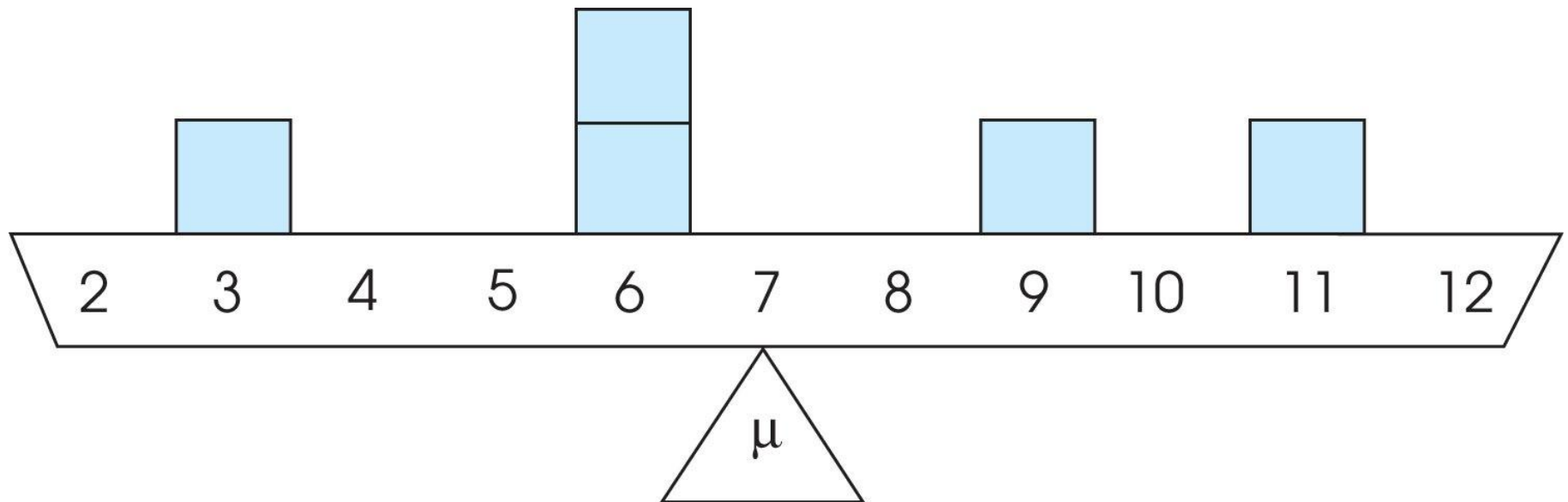
D

• $\Sigma X = 96$

Characteristics of the Mean

- Changing the value of a score changes the mean
- Introducing a new score or removing a score changes the mean (unless the score added or removed is exactly equal to the mean)
- Adding or subtracting a constant from each score changes the mean by the same constant
- Multiplying or dividing each score by a constant multiplies or divides the mean by that constant

Figure 3.3 – Mean is Highly Sensitive to Changes in Scores



Learning Check

A sample of $n = 7$ scores has $M = 5$. All of the scores are doubled. What is the new mean?

A

• $M = 5$

B

• $M = 10$

C

• $M = 25$

D

• More information is needed to compute M

3.3 The Median

- The median is the midpoint of the scores in a distribution when they are listed in order from smallest to largest
- The median divides the scores into two groups of equal size

Example 3.5

Locating the Median (odd n)

- Put scores in order
- Identify the “middle” score to find median

3 5 8 10 11

“Middle” score is 8 so median = 8

Example 3.6

Locating the Median (even n)

- Put scores in order
- Average middle pair to find median

1 1 4 5 7 9

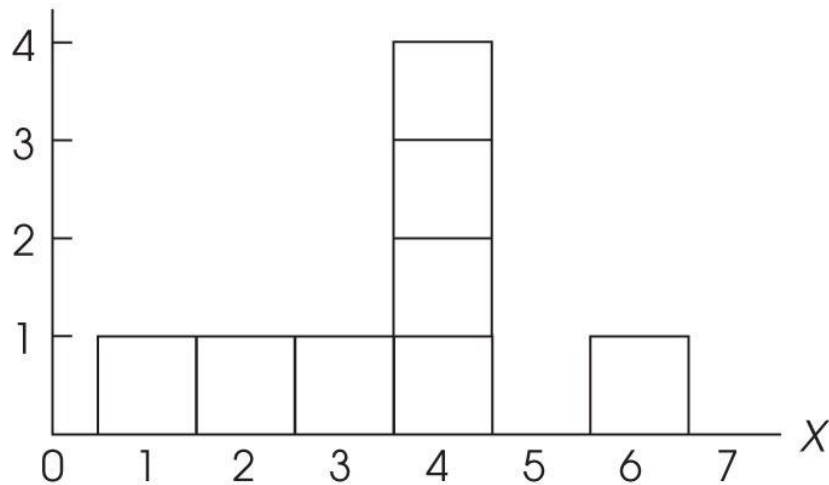
$$(4 + 5) / 2 = 4.5$$

The Precise Median for a Continuous Variable

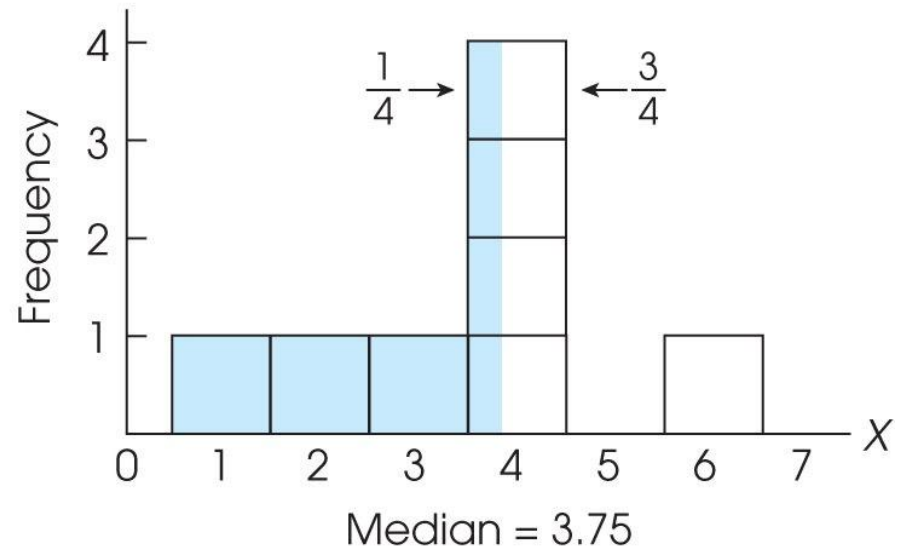
- A continuous variable can be infinitely divided
- The precise median is located in the interval defined by the real limits of the value.
- It is necessary to determine the fraction of the interval needed to divide the distribution exactly in half.
- $$\text{fraction} = \frac{\text{number needed to reach 50\%}}{\text{number in the interval}}$$

Figure 3.4 – Finding a Precise Median for a Continuous Variable

(a)



(b)



- The median is calculated using the formula for grouped data:
- Median= $L + ((N/2 - CF)/f) * w$

Where:

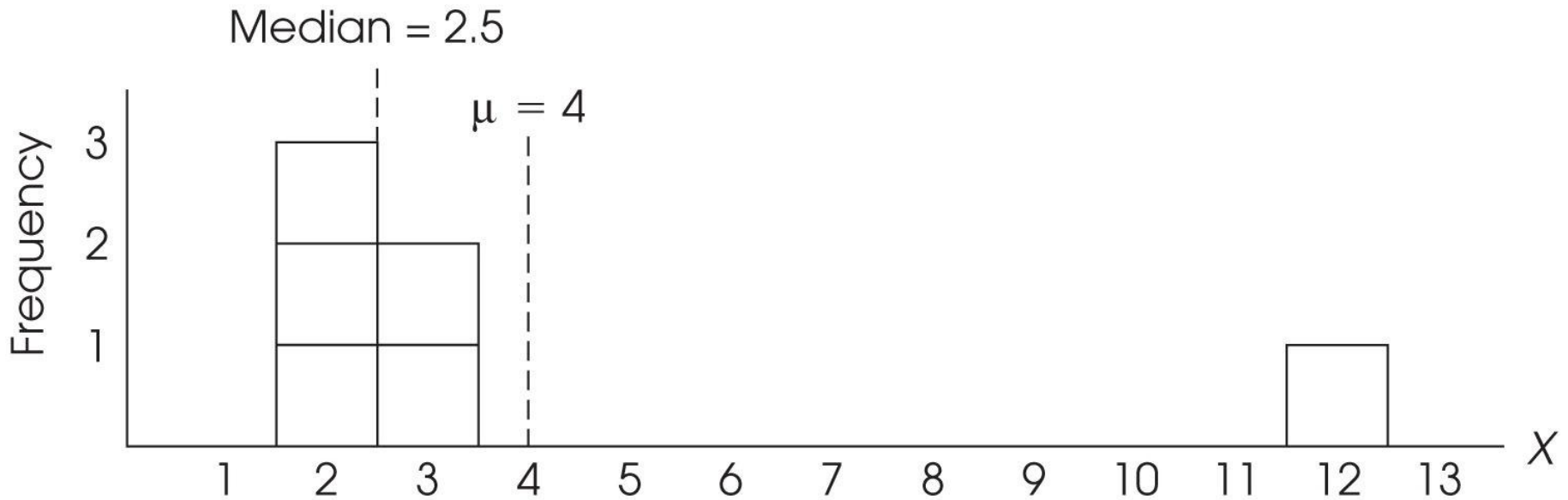
- L is the lower boundary of the median class, which is 3.5.
- N is the total frequency, which is 8.
- CF is the cumulative frequency of the class before the median class, which is 3 (f=1 for x=1, f=1 for x=2, and f=1 (x=3), then CF=3).
- f is the frequency of the median class, which is 4 (we have x=4 for 4 blocks).
- w is the width of the class, which is 1.
- Median= $3.5 + ((8/2 - 3)/4) \times 1 = 3.5 + ((4 - 3)/4) \times 1 = 3.5 + 0.25 = 3.75$

Answer: The median is 3.75

Median, Mean, and “Middle”

- Mean is the balance point of a distribution
 - Defined by distances
 - Often is not the midpoint of the scores
- Median is the midpoint of a distribution
 - Defined by number of scores
 - Often is not the balance point of the scores
- Both measure central tendency, using two different concepts of “middle”

Figure 3.5



Learning Check

- Decide if each of the following statements is True or False.

T/F

- It is possible for more than 50% of the scores in a distribution to have values above the mean

T/F

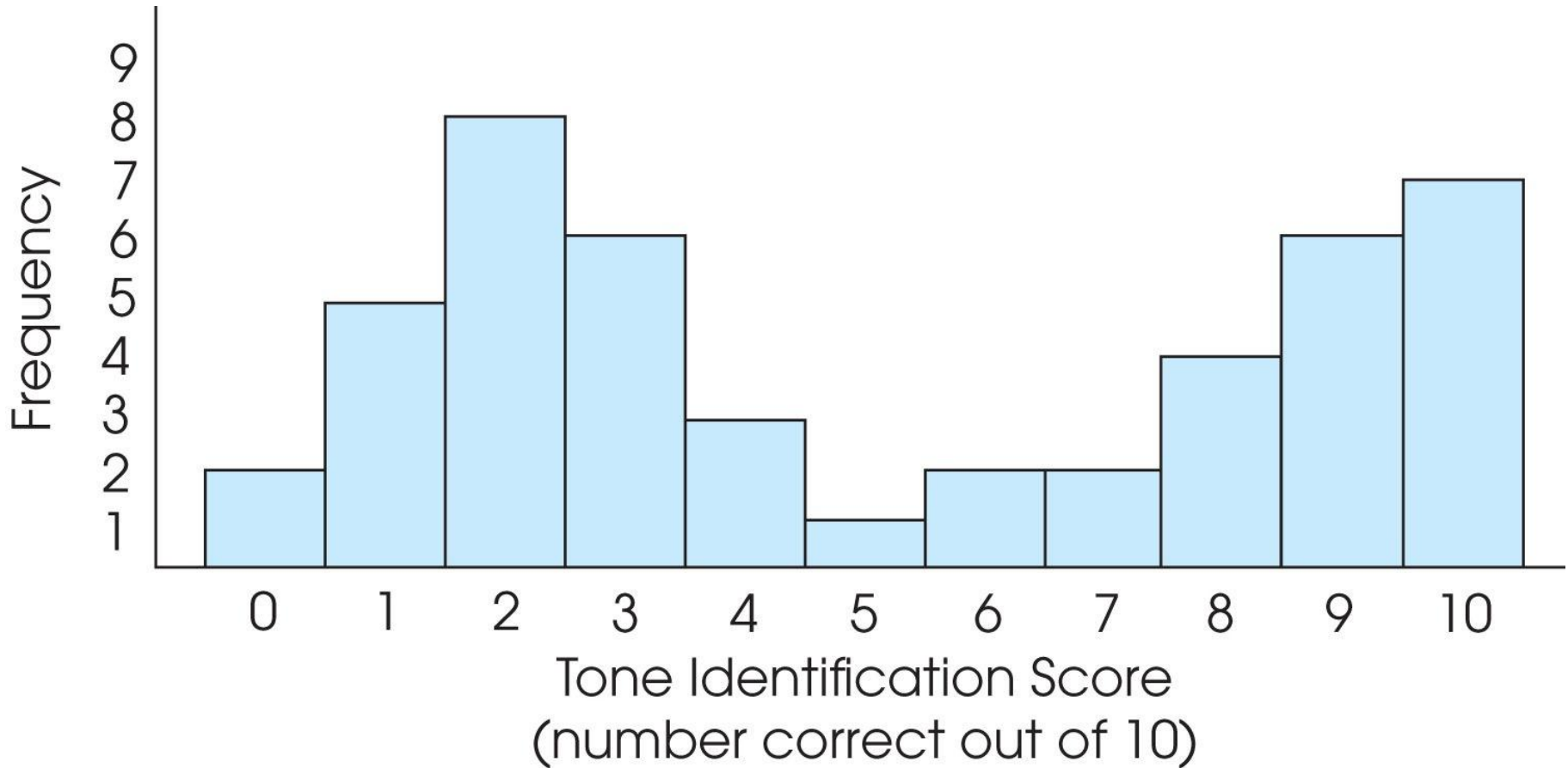
- It is possible for more than 50% of the scores in a distribution to have values above the median

3.4 The Mode

- The mode is the score or category that has the greatest frequency of any score in the frequency distribution
 - Can be used with any scale of measurement
 - Corresponds to an actual score in the data
- It is possible to have more than one mode

Figure 3.6

Bimodal Distribution



3.5 Selecting a Measure of Central Tendency

Measure of Central Tendency	Appropriate to choose if ...	Could be misleading if...
Mean	• You can calculate $\sum X$ • You know the value of every score	• Extreme scores • Skewed distribution • Undetermined values • Open-ended distribution • Ordinal scale • Nominal scale
Median	• Extreme scores • Skewed distribution • Undetermined values • Open-ended distribution • Ordinal scale	• Nominal scale
Mode	• Nominal scales • Discrete variables • Describing shape	• Interval or ratio data, except to accompany mean or median

Figure 3.7

Showing Large Gaps in Data

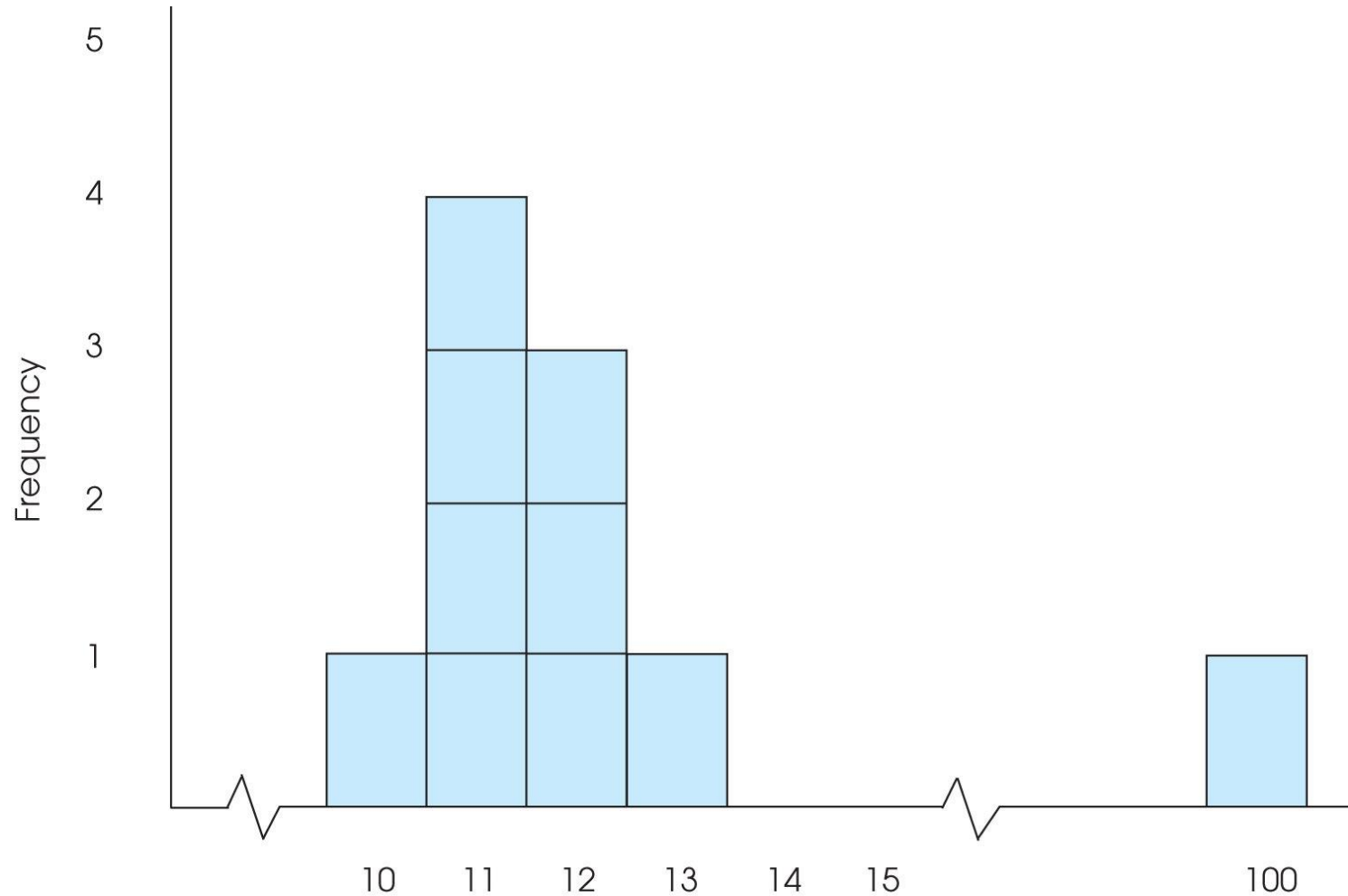


Figure 3.8

Means or Medians in a Line Graph

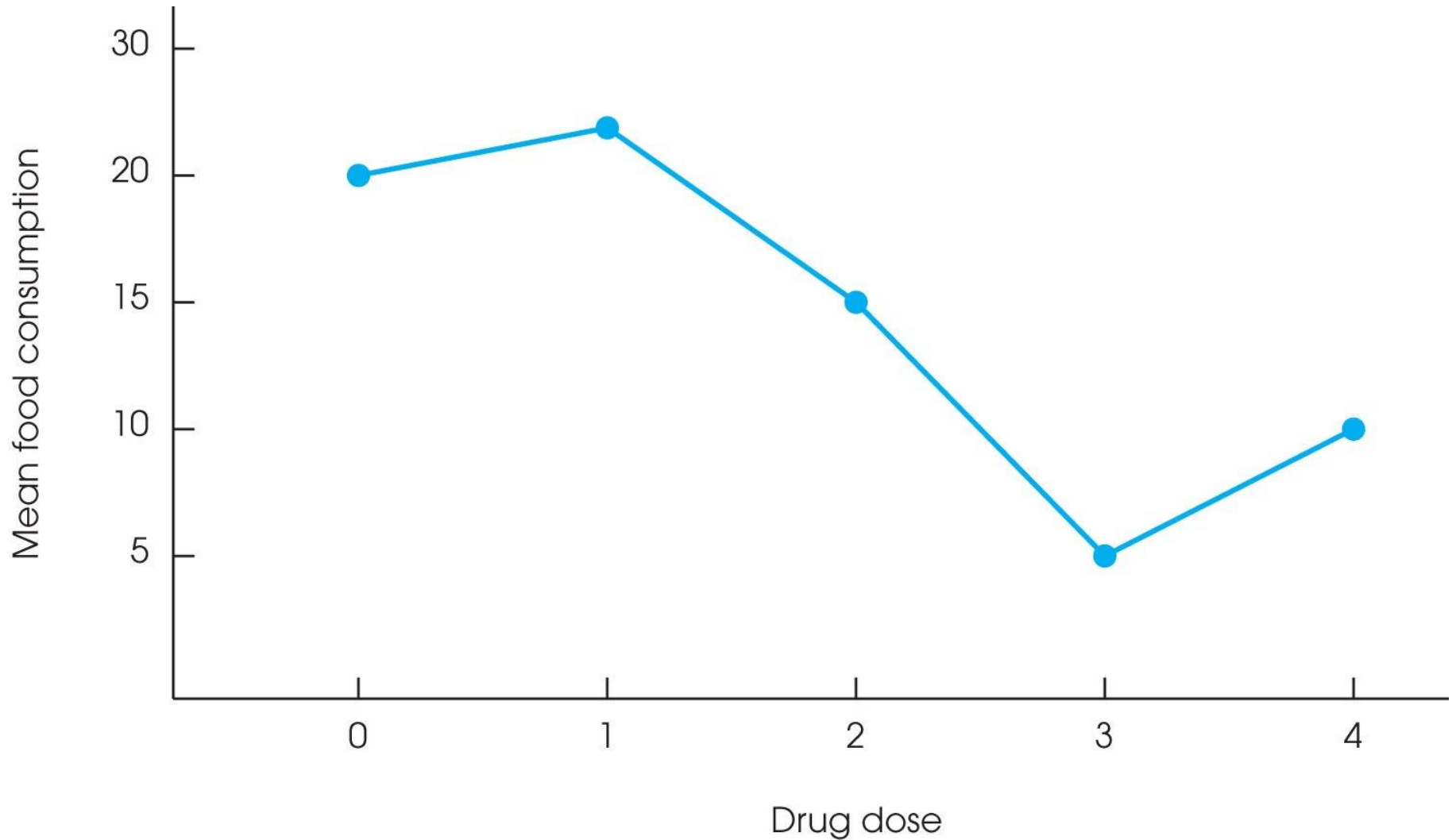
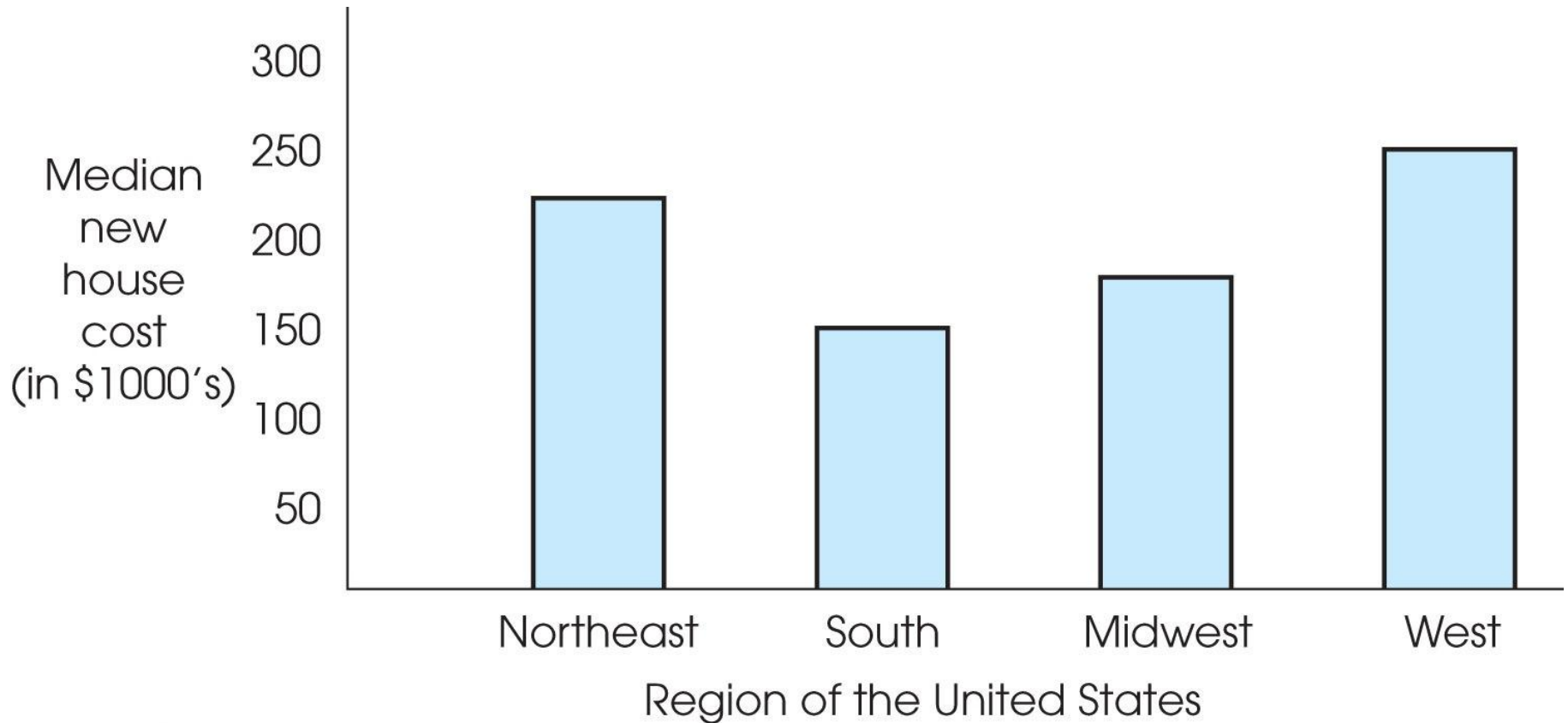


Figure 3.9

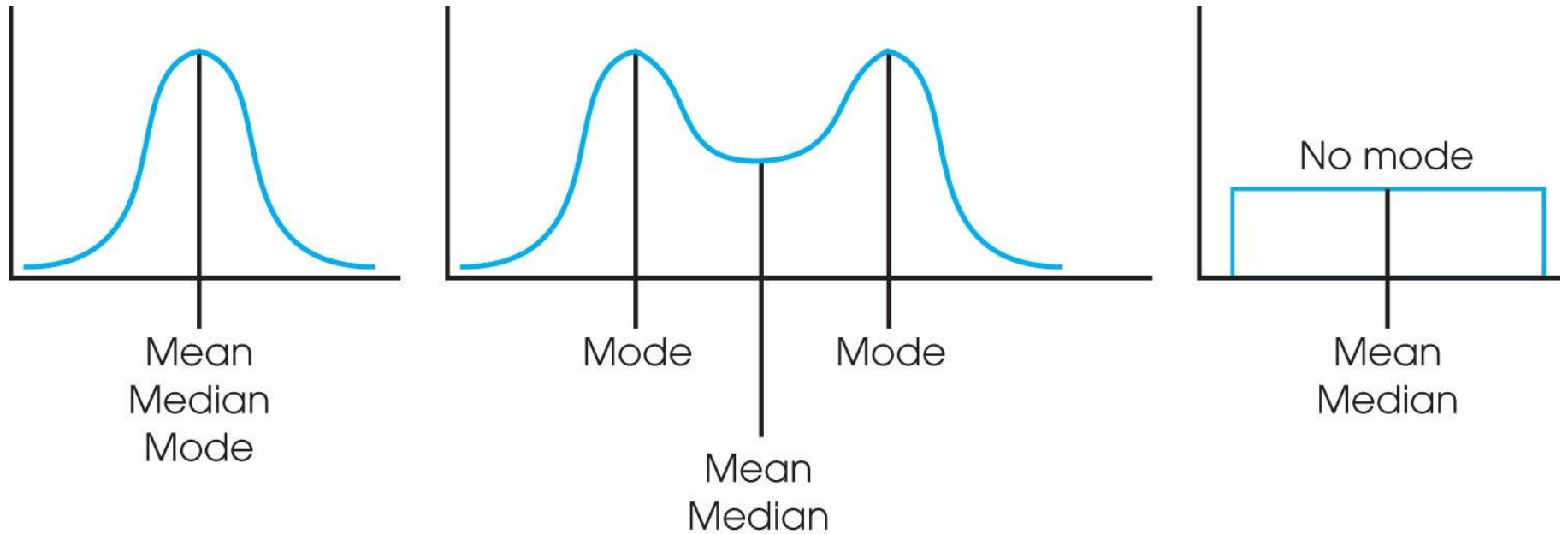
Means or Medians in a Bar Graph



3.6 Central Tendency and the Shape of the Distribution

- Symmetrical distributions
 - Mean and median have same value
 - If exactly one mode, it has same value as the mean and the median
 - Distribution may have more than one mode, or no mode at all

Figure 3.10

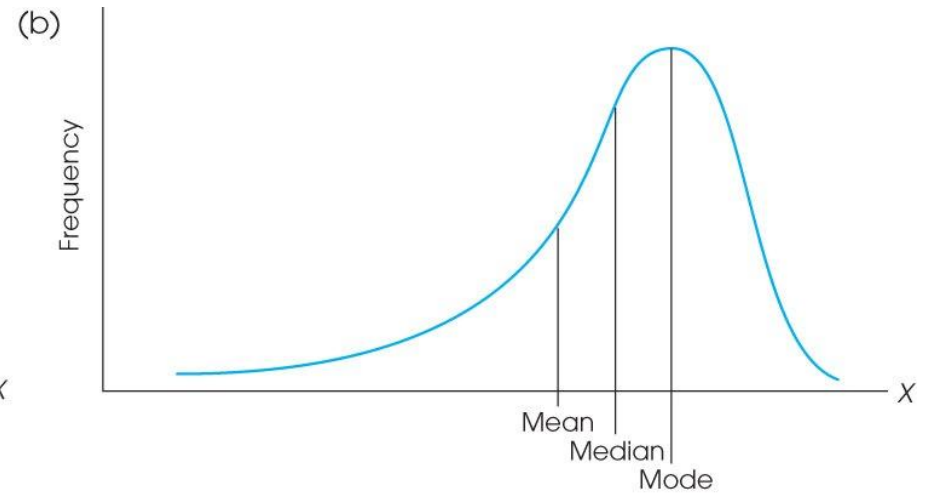
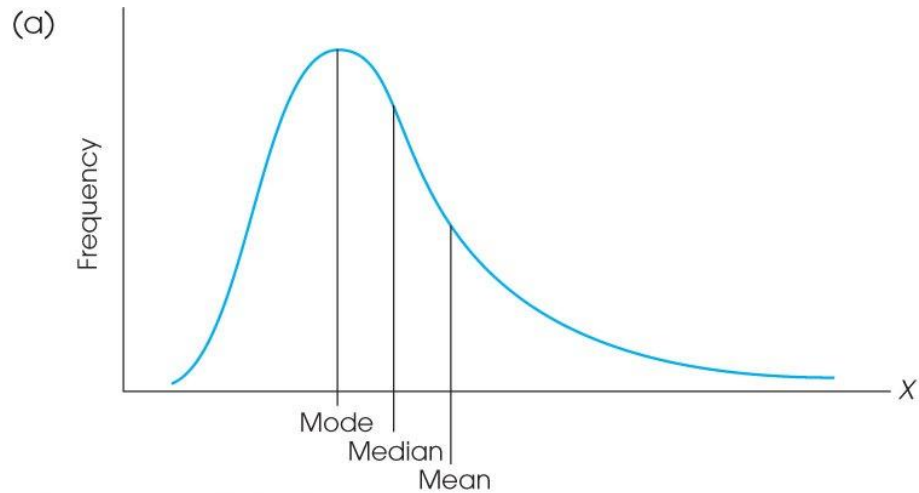


Central Tendency in Skewed Distributions

- Mean, influenced by extreme scores, is found far toward the long tail (positive or negative)
- Median, in order to divide scores in half, is found toward the long tail, but not as far as the mean
- Mode is found near the short tail.
- If $\text{Mean} - \text{Median} > 0$, the distribution is positively skewed.
- If $\text{Mean} - \text{Median} < 0$, the distribution is negatively skewed

Figure 3.11

Skewed Distributions



Learning Check

- A distribution of scores shows Mean = 31 and Median = 43. This distribution is probably

A

- Positively skewed

B

- Negatively skewed

C

- Bimodal

D

- Open-ended